

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Muthu Krishna K (192511037) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Muthu Krishna K.
Signature of the student:

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO-2 - ASSESSMENT-1
Analytical Problems.

Question no: 2.

Subnet Mask = 255. 255. 255. 192 = /26

Given.

* System 1 IP Address = 192.168.1.130

* System 2 IP Address = 192.168.1.190

* Subnet Mask = 255. 255. 255. 192 (/26)

Subnet Mask = /26

Host Bits = $32 - 26 = 6$

usable hosts = $2^6 - 2 = 62$ hosts.

Block Size = $256 - 192 = 64$.

Subnet ranges.

Subnet	Range.
192.168.1.0/26	0 - 63
192.168.1.64/26	64 - 127
192.168.1.128/26	128 - 191
192.168.1.192/26	192 - 255

Analyze System 1.

IP Address = 192.168.1.130

130 falls within 128 - 191.

Therefore,

Step 1: Allocate HR Department

Mask = /25.

Host bits = 7.

usable Hosts = $2^7 - 2 = 126$.

Assigned Subnet:

Network Address = 172.16.0.0/25

Host Range.

172.16.0.1 - 172.16.0.126.

Broadcast Address:

172.16.0.127.

Step 2: Allocate Sales Department

Next available address = 172.16.0.128.

Mask = /26.

usable Hosts = $2^6 - 2 = 62$.

Network Address = 172.16.0.128/26.

Host Range.

172.16.0.129 - 172.16.0.190.

Broadcast Address:

172.16.0.191.

Parameter	System 1	System 2
IP Address	192.168.1.130	192.168.1.190
Network Address	192.168.1.128	192.168.1.128
Host Range	192.168.1.129 - 192.168.1.190	192.168.1.129 - 192.168.1.190
valid Host	yes	yes

Both Systems Belong to the Same Subnet (192.168.1.128/26). Therefore, they can communicate directly without requiring a router. Each Subnet provides 62 usable host addresses.

Question no: 3
Department-wise Subnet Allocation using VLSM.

Given.

Network Block = 172.16.0.0/16

Department

HR

Sales.

IT

Admin.

Mask.

/25

/26

/27

/28

Step: 3: Allocate IT Department.

Next available address = 172.16.0.192
Mask = /27.

Usable Host = $2^5 - 2 = 30$.

Network Address = 172.16.0.192/27.

Host Range:

172.16.0.193 - 172.16.0.222.

Broadcast Address:

172.16.0.223.

Step: 4 Allocate Admin Department.

Next available address = 172.16.0.224

Mask = /28

Usable Host = $2^4 - 2 = 14$.

Network Address = 172.16.0.224/28.

Host Range:

172.16.0.225 - 172.16.0.238.

Broadcast Address:

172.16.0.239.

Step 5: Determine Remaining unused Range.

Last utilized address:

172.16.0.239.

Next available address:

172.16.0.240.

Remaining unused range:

172.16.0.240 - 172.16.255.255.

